AI TOOL PROMPT ENGINEERING REPORT

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CPEng 2-A

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I. AI TOOL OVERVIEW

Name of AI: Flowith

Purpose: To generate images/ media for users to use.

Why: I wanted to experience how prompts really affect outputs of AI generated content.

II. ORIGINAL PROMPT

"A tiny, high-tech laboratory hidden inside a hollowed-out coconut on a tropical beach, run by a very serious squirrel wearing tiny safety goggles."

Output:

Simple Generation of said Image

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III. AI-ASSISTED PROMPT

Using ChatGPT to Modify the Prompt:

Make this prompt better for Flowith
"A tiny, high-tech laboratory hidden inside a hollowed-out coconut on a tropical beach, run by a very serious squirrel wearing tiny safety goggles."

Here's a more vivid, Flowith-optimized version of your prompt with stronger visual detail, mood, and composition:

Enhanced Prompt:

A hyper-detailed miniature scene of a high-tech laboratory built inside a hollowed-out coconut on a sunlit tropical beach. The coconut interior is transformed into a futuristic lab with glowing holographic screens, glass vials, and intricate machinery. At the center, a very serious squirrel scientist wearing tiny safety goggles carefully conducts an experiment, surrounded by bubbling chemicals and blinking instruments. Soft golden sunlight filters through the coconut opening, contrasting with the cool neon lights inside. Ultra-realistic textures, cinematic lighting, shallow depth of field, 8K detail, whimsical yet grounded in realism.

If you want, ↳ I can tailor it for a specific style (Pixar, anime, photorealistic, isometric, etc.).

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IV. COMPARISON AND ANALYSIS

Observed Differences:
The first image (before improvement) already captures the core idea: a squirrel scientist working inside a coconut laboratory on a beach. However, it feels more like a straightforward interpretation of the prompt. The lab setup is somewhat generic, and while the elements (test tubes, machines, goggles) are present, they lack a strong sense of atmosphere or narrative depth. The lighting is even and natural, but not particularly dramatic, and the overall composition feels slightly static.

In contrast, the second image (after improvement) is significantly more immersive and cinematic. The laboratory appears more advanced, with glowing holographic displays, richer instrumentation, and a more intricate arrangement of objects. The lighting is more dynamic, with warm sunlight from the beach contrasting against cool neon hues inside the coconut. The squirrel appears more engaged in a specific action, enhancing the storytelling aspect. The scene also benefits from better depth of field and a more deliberate composition.

Which Output Was More Useful or Accurate? Why?
The improved output is more useful and accurate because it aligns more closely with the intended vision of a “high-tech” and “whimsical yet realistic” lab. It better captures the fusion of natural and futuristic elements, which was only partially realized in the first image. The added visual complexity and mood make it more suitable for creative, professional, or storytelling purposes.

Impact of the Improved Prompt:

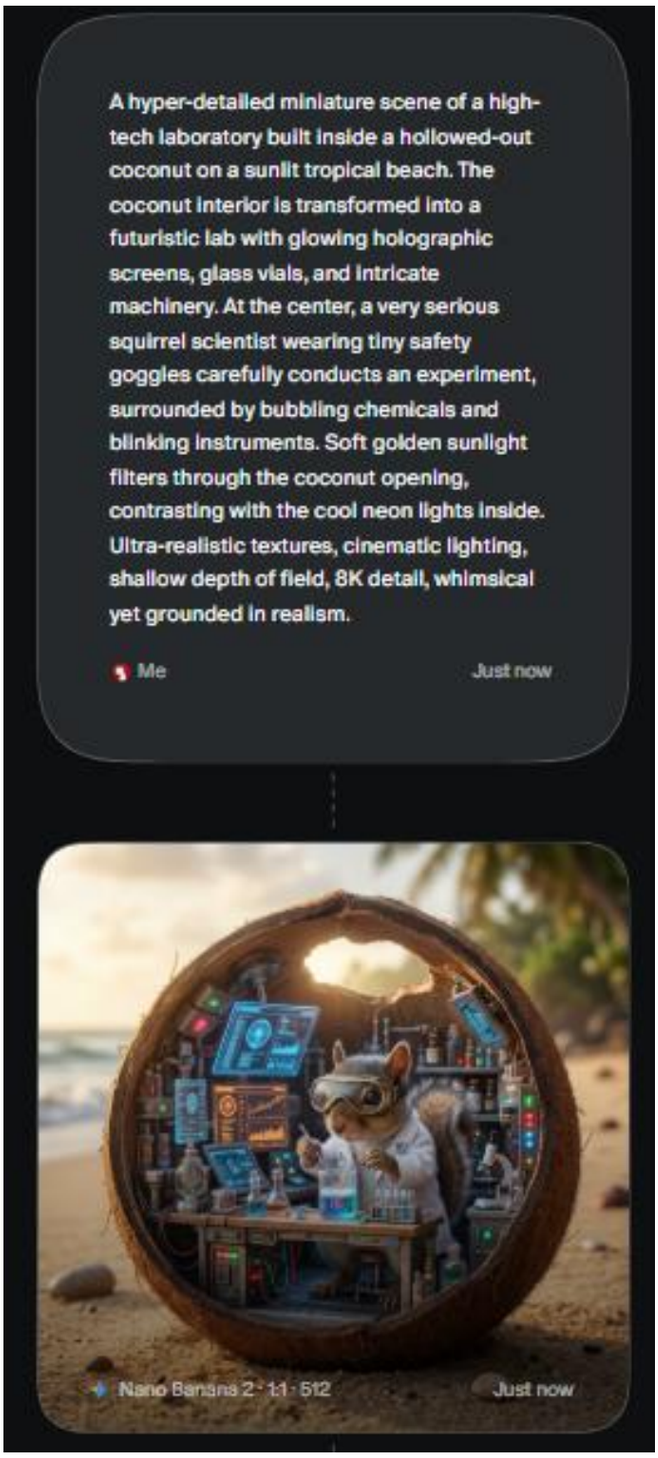
- **Clarity:**
  The enhanced prompt provided clearer guidance by specifying elements like “glowing holographic screens,” “cinematic lighting,” and “ultra-realistic textures.” This reduced ambiguity and helped the model prioritize key visual features.
- **Depth:**
  The second image demonstrates greater depth both visually and conceptually. Details such as layered lighting, intricate machinery, and varied textures create a richer scene. The squirrel’s action (actively experimenting) adds narrative depth.
- **Relevance:**
  Every element in the improved output feels intentional and aligned with the prompt. The futuristic lab components, lighting style, and composition all directly reflect the added descriptive phrases.

Specific Changes That Led to Better Results:
Key improvements included:

- Adding **adjectives and qualifiers** (e.g., “hyper-detailed,” “futuristic,” “intricate machinery”)
- Specifying **lighting conditions** (“cinematic lighting,” “soft golden sunlight,” “cool neon lights”)
- Including **camera/visual cues** (“shallow depth of field,” “8K detail”)
- Enhancing **narrative context** (the squirrel actively conducting experiments)
- Introducing **contrast** (natural beach vs. high-tech interior)

What I Learned About Prompt Engineering:
This comparison highlights that specificity and intentionality are crucial. A good prompt doesn’t just describe objects—it defines mood, style, lighting, and composition. Including cinematic and technical details significantly improves output quality. Additionally, guiding the model with contrasts and storytelling elements leads to more engaging and visually compelling results.

A tiny, high-tech laboratory hidden inside a hollowed-out coconut on a tropical beach, run by a very serious squirrel wearing tiny safety goggles.
Me
Just now
Nano Banana 2 · 1:1 · 512
Just now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